ENGINEERING RESEARCH CORPORATION LLC

# Hardware and Firmware Validation Maturity Model

What does capable hardware and firmware validation look like, and how would you know it if you saw it? A five-level, evidence-gated framework covering 17 practice domains — from manually executed bring-up and reactive debugging to requirements-traceable, automated, risk-driven, continuously measured validation — with scoring rules, evidence instruments, and anti-precision constraints.

**ERC Flagship Report ERC-RPT-2026-003**

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The model is constructed from documented standards and engineering literature with clearly labeled ERC analysis; structured practitioner-interview validation is scheduled before final release. All instruments (evidence-request list, interview guide, assessment workbook, heat-map template) are published so any organization can perform the same assessment without ERC. The report is a methodology, not a certification scheme.

# Document control and disclosures

## Document control

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## Disclosures

**Sponsor or client:** None. ERC-initiated and self-funded; future sponsorship will be disclosed on every distributed copy.

**Conflicts of interest:** This report describes a methodology ERC intends to use in paid assessments. That commercial interest is disclosed here and in the final release. The model, scoring rules, and instruments are published openly so any organization can perform the same assessment without ERC. ERC will not rate any organization publicly.

**Basis of the model:** The model synthesizes published standards and literature with clearly labeled ERC analysis. Founder-level practitioner experience informs domain selection and failure-mode lists; such inputs are labeled *Interpretation* wherever they appear, and the scheduled interview validation may revise the model.

**Editorial independence:** Findings are reported regardless of commercial effect.

**Product-name disclaimer:** Tool and standard names are designations of their respective owners, used for identification only. The model explicitly does not score tool ownership.

**Use of AI disclosure:** This report was drafted with AI writing assistance under ERC staff direction. Every external claim was verified against the cited source before inclusion; all

model-specific content is labeled as ERC analysis.

**Not legal or certification advice:** This model supports internal improvement and due-diligence conversations. It is not a certification scheme, does not replace ISO audits or regulatory submissions, and confers no compliance status.

# Abstract

Hardware and firmware validation determines whether defects are found before or after shipment, how quickly they can be reproduced, and whether recurrence is prevented — yet organizations rarely have a neutral way to describe how capable their validation actually is. This report defines a five-level Hardware and Firmware Validation Maturity Model (ERC-HVM 1.0) covering 17 practice domains, from requirements traceability, test planning, board bring-up, and instrumentation through regression automation, configuration control, fault injection, security testing, test-data governance, laboratory reproducibility, release criteria, and field-feedback integration. The model is constructed from documented engineering practices and standards — CMMI and TMMi maturity structures, NIST SP 800-193 platform-firmware resiliency guidance, NIST SP 800-218 secure-development practices, NASA bidirectional-traceability requirements, ISO/IEC/IEEE 29119 test documentation, and ISO/IEC 17025 laboratory competence — plus clearly labeled ERC analysis. Every domain criterion specifies the expected practice, required artifact, measurement method, responsible role, and common failure modes. Scoring assigns a level only when objective evidence demonstrates repeatability and demonstrated influence on release decisions; tool ownership alone never earns credit. The report separates process maturity from product quality using published defect-removal-efficiency data, and provides decision instruments, an evidence-request list, an interview guide, and anti-precision scoring rules. Principal implication: maturity is measurable, improvable in ranked steps, and directly connected to schedule risk, escaped defects, rework, coverage, and release confidence.

## Keywords

hardware validation; firmware validation; maturity model; requirements traceability; regression automation; board bring-up; fault injection; configuration control; defect removal efficiency; release criteria; TEVV; due diligence

## ONE-PAGE DECISION SUMMARY — for engineering directors and due-diligence teams

**The decision.** How capable is this organization at finding, reproducing, prioritizing, and preventing hardware and firmware defects — and what should be improved first, or asked first in diligence?

**What the evidence base establishes.** (1) Maturity models are credible only when evidence-gated: CMMI and TMMi define levels as evolutionary plateaus demonstrated through practice, not tool ownership. (2) Maturity moves outcomes: a published case study applying TMMi levels 2–4 raised test-case effectiveness from 51% to 84% and defect quality from 69% to 89%. (3) What immaturity costs: typical defect removal efficiency (DRE) is near 85%; individual test stages remove only 25–42% of defects each; combined prevention, pre-test removal, and formal testing can top 99%; and a NASA study measured requirements errors growing 3–8x in cost by the design phase and escalating thereafter. (4) The underlying practices are already specified by authorities: NASA requires bidirectional traceability; NIST SP 800-193 defines firmware protect/detect/recover with cryptographic verification; ISO/IEC 17025 defines laboratory result validity.

**The model.** Five levels (L1 Initial, L2 Managed, L3 Defined, L4 Quantitatively Managed, L5 Optimizing) across 17 domains, from board bring-up and instrumentation to fault injection, security testing, release criteria, and field-feedback integration. A level is earned only when the practice is documented, the artifact is current, the measurement is reproducible, a role owns it, and the evidence demonstrably influenced a release decision. Tool ownership earns nothing. Output is a domain heat map — never a decimal score.

**Confidence.** High in the structural elements (anchored to published frameworks and standards); moderate on any specific organization placement until an assessment runs; the model itself will be revised after scheduled practitioner interviews.

**Limitations.** Not yet interview-validated; ordinal levels resist false precision by design; assessment effort is real (a staged five-domain "vital signs" option is provided); DRE evidence depends on field visibility.

# Executive summary

## Decision problem

Engineering directors, investors, acquirers, and due-diligence teams need to know not whether a hardware or firmware organization has defects — every organization does — but whether that organization finds, reproduces, prioritizes, and prevents recurrence of defects reliably. A mature organization can still ship a serious defect; an immature one cannot say when it was introduced, cannot reproduce it on demand, and will ship it again. This report provides the instrument for answering the capability question with evidence.

## Key findings

**1. Evidence-gated maturity works; tool-gated maturity does not measure anything.** CMMI defines five maturity levels — Initial, Managed, Defined, Quantitatively Managed, Optimizing — each a well-defined evolutionary plateau of process improvement [1][2][20]. TMMi applies the same logic to test processes with five levels: Initial, Managed, Defined, Measured, Optimization [3][4]. Neither framework awards levels for owning tools; both require demonstrated practice. ERC-HVM inherits this logic.

**2. Maturity measurably moves defect outcomes.** A peer-reviewed industrial case study applying TMMi levels 2–4 with risk-based test techniques reported test-case effectiveness rising from 51% to 84% and a defect-quality measure rising from 69% to 89% [16]. This is a single case study, labeled as such — but it is the kind of outcome the maturity investment is supposed to buy.

**3. Defect-removal efficiency quantifies the cost of weak validation.** Published benchmark analyses report typical DRE near 85%, with unit test alone 25–38% efficient, function test around 40%, and regression around 35% in documented cases; design and code inspections often exceed 85% efficiency; and combined prevention, pre-test removal, and formal testing can top 99% [11][12][13][14][19]. A published 37-project dataset showed DRE of 72% with 28 defects detected after delivery [17]. DRE is computable in hardware and firmware programs and is a Level-4 criterion in this model.

**4. Late discovery is expensive on a measured curve.** NASA measured the cost of fixing a requirements-phase error at 1 unit when found during requirements, 3–8 units when found during design, and escalating through later phases [15]. For hardware the curve is steeper still: a board defect found after fabrication invokes respin; found after shipment, it invokes field returns and brand damage.

**5. The practices are already specified by authorities — the model measures adoption.** NASA NPR 7150.2D requires bidirectional traceability between requirements, design, code, and test procedures [7]. NIST SP 800-193 defines platform-firmware resiliency as protect, detect, and recover, with cryptographic verification of firmware code before execution [5]. NIST SP 800-218 defines secure-development practices including root-cause prevention of vulnerability recurrence [6]. ISO/IEC/IEEE 29119-3 defines the test-documentation set [8]; ISO/IEC 17025 defines when laboratory results are valid and

traceable [10]. ERC-HVM organizes these into levels an organization can act on.

## Confidence and limitations

Confidence is high that the structural elements — five levels, 17 domains, evidence gating, anti-precision scoring — reflect established, peer-documented practice. Confidence is moderate for any specific organization placement until an assessment executes with the published workbook and interview guide. Major limitations: the model is not yet interview-validated; ordinal levels must never be averaged into decimals; and DRE evidence depends on field visibility. Section 9 addresses each.

## Recommendations (preview)

Assess before investing: baseline the five vital-sign domains first (traceability, regression automation, configuration control, release criteria, defect management). Gate every improvement on artifacts that demonstrably influence release decisions. Never purchase maturity in the form of tools. Section 10 provides the sequenced roadmap; Section 7 provides the scoring rules that keep the assessment honest.

## Reader guide

The **one-page decision summary** (previous page) serves executives and diligence teams. **Sections 1–3** establish purpose, questions, and scope. **Section 4** documents how the model was constructed and how the supporting evidence was selected. **Section 5** maps every domain to its anchor standards and presents the outcome evidence. **Section 6** presents the model: levels, domains, the maturity matrix, and failure modes. **Sections 7–8** provide scoring rules and the maturity-versus-quality distinction. **Sections 9–10** cover limitations and the adoption roadmap. **Appendices A–J** hold the evidence-request list, interview guide, three fully specified exemplar domains, the heat-map template, the accessibility statement, assessment operations, full Level-3 criteria for all 17 domains, tailoring guidance, the scoring quick-reference card, and the Level-2 floor criteria.

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# 1. Purpose and decision context

## 1.1 Why this report exists

**Conclusion first.** "How good is your validation?" is answerable — but only with a structured model, defined evidence, and scoring rules that resist inflation. This report provides all three, openly.

This is ERC's third flagship report and its closest match to the company core practice: validation, automation, server administration, firmware, and board-level testing. The deliverable is a reusable methodology — a maturity model an organization can self-apply, a due-diligence team can apply to a target, and ERC can apply in engagements — with every instrument published so results are comparable and auditable. That openness is deliberate: a maturity model sold as a black box cannot be audited by the organizations it judges.

## 1.2 Primary audiences and decisions

**Table 1. Primary audiences and the decisions this report supports**

| Audience | Decision this report supports |
|---|---|
| Engineering directors (semiconductor, board and system, embedded, OEM) | Where to invest in validation capability next; what "good enough to ship" should mean |
| Firmware teams | Which practices to adopt, in what order, with defined artifacts |
| Technical due-diligence groups (investors, acquirers) | A structured, evidence-based view of a target validation capability and its schedule and quality risk |
| Laboratories | A path toward ISO/IEC 17025-style result validity and reproducibility |
| Workforce-development programs | A skills-and-practice curriculum grounded in real evidence requirements |

**Executive language standard.** Executive sections discuss schedule risk, escaped defects, rework, coverage, and release confidence. Engineering sections use precise terminology: configurations, tolerances, reproducibility, instrumentation, and uncertainty. Both registers appear, separated by section, per the ERC audience standard.

## 1.3 What makes a maturity model credible

Three properties, all enforced here: (1) **Levels are earned by evidence, not purchased** — consistent with CMMI and TMMi definitions of maturity as demonstrated practice [1][3]; (2) **criteria are specified to the artifact** — an assessor can reproduce the measurement from what the organization produces; (3) **scores stay ordinal** — the model forbids decimal aggregates that would imply measurement precision the method does not have.

# 2. Research questions

**Governing point.** Eight questions structure the report. Each is answered or explicitly left unresolved.

**Table 2. Research questions and where they are addressed**

| # | Research question | Addressed in |
|---|---|---|
| RQ1 | What does capable hardware and firmware validation look like at progressively mature organizations? | Section 6 |
| RQ2 | Which domains must the model cover, and how were they selected and bounded? | Sections 3, 6.2 |
| RQ3 | What objective evidence distinguishes each level in each domain — and what failure modes accompany each? | Sections 6.4, 7.2, Appendix C |
| RQ4 | How can maturity be scored without tool-gaming or false precision? | Section 7 |
| RQ5 | How does process maturity relate to product quality, and how are the two kept separate? | Section 8 |
| RQ6 | Which standards and literature already define the underlying practices? | Section 5 |
| RQ7 | How should improvement be sequenced, and what does a remediation roadmap look like? | Section 10 |
| RQ8 | What limits the model validity, and what validation remains? | Section 9 |

# 3. Scope and definitions

## 3.1 Scope

**In scope:** validation and verification practices for hardware (boards, systems, semiconductors) and the firmware that runs on them, from first power-on through release and field feedback, in any industry.

**Out of scope:** product certification regimes as such (UL, CE, FCC testing is acknowledged as context, not scored); software-only application testing; manufacturing yield engineering except where factory test data feeds the domains; and any claim about specific vendors products.

## 3.2 The 17 domains

The model covers 17 domains: requirements traceability; test planning; board bring-up; instrumentation; firmware qualification; regression automation; configuration control; defect taxonomy and management; observability; fault injection; stress testing; interoperability; security testing; test-data governance; laboratory reproducibility; release criteria; and field-feedback integration. Selection logic and bounding are in Section 6.2.

## 3.3 Definitions

**Table 3. Key definitions**

| Term | Definition |
|---|---|
| Validation | Confirmation, through objective evidence, that requirements for a specific intended use are fulfilled (aligned with ISO/IEC 25000 vocabulary as cited by NASA) |
| Verification | Confirmation that specified requirements are fulfilled by the work product |
| Maturity level | A well-defined evolutionary plateau of process improvement (CMMI usage) |
| Escaped defect | A defect discovered by a party other than the developing organization after handoff |

| Term | Definition |
|---|---|
| Defect removal efficiency (DRE) | Defects found and fixed before release divided by defects present; requires post-release visibility, hence a Level-4-and-above criterion |
| Regression evidence | A test-result artifact whose preconditions, firmware and driver versions, environmental settings, and pass/fail thresholds are recorded and machine-readable |
| Bring-up | First power-on and incremental enabling of a new board or system until stable and testable |
| Configuration record | The complete statement of firmware, driver, fixture, instrument, and environment versions in effect for a test session |
| Instrumentation | The calibrated instruments and measurement procedures used to observe the device under test |
| Fault injection | Deliberate introduction of faults (voltage, clock, bus, software) to verify detection and recovery behavior |
| Test-data governance | Defined schema, retention, integrity, access, and quality rules for test results |
| Laboratory reproducibility | Ability to re-obtain a result within stated tolerance under the recorded environment (ISO/IEC 17025 alignment) |

# 4. Methodology: how the model was constructed

**Governing point.** The model is constructed, not invented: each domain and criterion traces to at least one published standard or authoritative source, or is explicitly labeled ERC analysis. Nothing rests on tool ownership.

## 4.1 Construction protocol

**Step 1 — Anchor levels in published maturity structures.** Level names and progression logic adopt the CMMI and TMMi five-level structures, which define maturity as an evolutionary plateau demonstrated through practice [1][3][4]. ERC does not resell or certify either model; the vocabulary is cited, not copied wholesale.

**Step 2 — Derive domains from documented practice.** Domain candidates came from NIST SP 800-193 firmware-resiliency guidance, NIST SP 800-218 SSDF practices, NASA NPR 7150.2D traceability and assurance requirements, ISO/IEC/IEEE 29119-3 test documentation, ISO/IEC 17025 laboratory requirements, published DRE analyses, and ERC practitioner experience (labeled as ERC analysis).

**Step 3 — Specify criteria with five mandatory fields.** Every criterion states: expected practice, required artifact, measurement method, responsible role, and common failure modes. Criteria lacking any field were revised or cut.

**Step 4 — Gate every level on evidence and influence.** A level is earned only when artifacts exist, are current, and demonstrably influence release decisions. Appendix A operationalizes this with an evidence-request list.

**Step 5 — Separate process maturity from product quality.** DRE, escape counts, and rework hours are product measures (Section 8); they are inputs to Level-4/5 criteria, never

substitutes for process evidence.

**Step 6 — Practitioner validation (scheduled).** Before final release, ERC will conduct structured interviews with at least ten experienced practitioners across at least four industries, scoring agreement between model criteria and practitioner-confirmed practice; results and revisions will be published. No interview claims are made in this draft.

## 4.2 Evidence-selection protocol for the supporting audit

**Search dates and coverage:** October 20–27, 2026. Sources: NIST CSRC and NTRS; ISACA and CMMI materials; TMMi Foundation; ISO and ANSI catalogs; ACM, IEEE, MDPI, and Frontiers publications; NASA technical reports.

**Search concepts:** maturity model test process; CMMI maturity levels; TMMi levels; firmware resiliency guidelines; secure development framework; bidirectional traceability; test documentation standard; laboratory competence; defect removal efficiency; cost of late defect discovery; electronics production escape risk.

**Inclusion criteria:** official standard or framework publication; peer-reviewed study with method disclosure; named-authority report with data. **Exclusion criteria:** vendor marketing; maturity-model marketing pages without published criteria; unverifiable anecdotes.

**Conflicts:** where frameworks disagree — for example, CMMI capability levels 0–3 per process area versus staged maturity levels 1–5 — both structures are noted and the ERC choice is justified in Section 5.3.

## 4.3 Calculation methods

No proprietary statistics are computed. Where this report cites DRE values, cost-escalation ratios, or case-study percentages, the values are quoted from the cited sources with their populations and periods. The assessment workbook computes only inspectable arithmetic: per-domain level assignment, gate checks, and heat-map shading — no hidden weighting.

# 5. Evidence base and standards landscape

**Lead point.** Every practice domain in the model maps to at least one authoritative source. This section shows the map, positions the model in the maturity-model landscape, and presents the outcome evidence that justifies evidence-gating.

## 5.1 Standards and guidance mapped to domains

**Table 4. Anchor standards and what they establish**

| Domain | Anchor standards and sources | What the anchor establishes |
|---|---|---|
| Requirements traceability | NASA NPR 7150.2D (SWE-052, SWE-071/072); ISO/IEC/IEEE 29119-3 | Bidirectional traceability between requirements, design, code, and test procedures is a required practice in mature assurance programs, not a nicety |

| Domain | Anchor standards and sources | What the anchor establishes |
|---|---|---|
| Test planning | ISO/IEC/IEEE 29119-3 (test plans, procedures, completion reports) | A standard test-documentation set with defined content |
| Firmware qualification | NIST SP 800-193; NIST SP 800-218 | Firmware code verified before execution (signature or hash); secure-development practices with root-cause prevention |
| Security testing | NIST SP 800-193 (protect, detect, recover); SP 800-218 | Resiliency against unauthorized firmware change is a defined engineering property, testable at each stage |
| Laboratory reproducibility | ISO/IEC 17025:2017 | Competence, impartiality, and consistent operation; valid, traceable results |
| Instrumentation and measurement | ISO/IEC 17025 (metrological traceability) | Measurement results must be traceable and valid for their intended use |
| Release criteria and field feedback | ISO/IEC/IEEE 29119-3 (test completion reports); SSDF root-cause practices | Completion evidence and post-release feedback feed prevention |
| All domains (structure) | CMMI; TMMi | The five-level evidence-gated progression logic |

*Note:* standards are cited by number and title; where ISO texts are paywalled, the citation identifies the standard and clause area without quoting the text.

## 5.2 Maturity-model landscape

**Table 5. Maturity-model landscape and positioning**

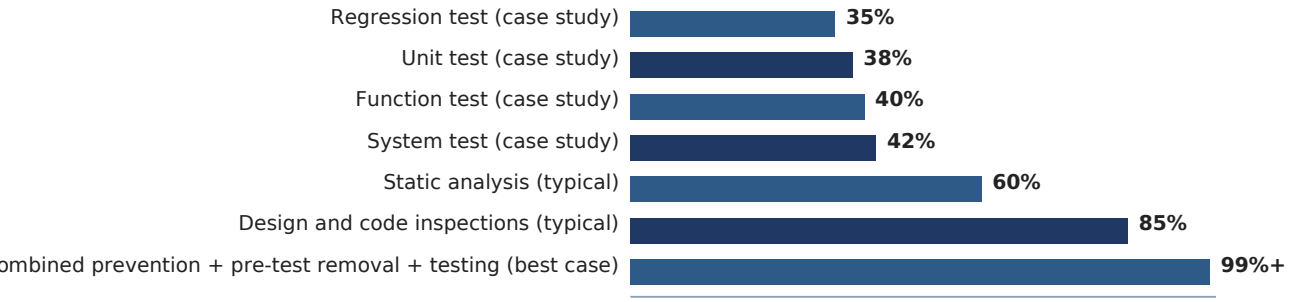
| Model | Scope | Five levels | Relationship to ERC-HVM |
|---|---|---|---|
| CMMI (ISACA) | Organization-wide development | Initial; Managed; Defined; Quantitatively Managed; Optimizing | Provides the level logic; CMMI also defines capability levels 0–3 per process area — ERC-HVM uses staged levels 1–5 for simplicity, stated as a deliberate choice |
| TMMi (TMMi Foundation) | Test process | Initial; Managed; Defined; Measured; Optimization | Closest prior model; software-test focused; ERC-HVM extends to hardware, board bring-up, instrumentation, firmware, and field feedback |
| ISO/IEC 17025 | Laboratories | Not a level model (requirements) | Supplies laboratory reproducibility and measurement-validity criteria embedded in ERC-HVM Level 3 and above |
| NIST SP 800-193 / 800-218 | Firmware resiliency; secure development | Not a level model (practice frameworks) | Supply the firmware and security-testing criteria embedded across levels |

## 5.3 Outcome evidence: why evidence-gated maturity matters

**Maturity moves outcomes (one measured case).** Applying TMMi levels 2–4 with risk-based techniques in an industrial setting raised test-case effectiveness from 51% to 84% and defect quality from 69% to 89% [16]. A single case study, cited as such.

**DRE benchmarks quantify weak validation.** Typical DRE is near 85%; unit test alone 25–38% efficient; design and code inspections often top 85%; combined prevention, pre-test removal, and formal testing can top 99% [11][12][13][14][19]. A 37-project dataset showed 72% DRE with 28 post-delivery defects [17].

**Figure 1. Published defect-removal efficiency by removal operation (percent of defects removed, as reported by the cited analyses)**




Source: Jones DRE analyses via PPI and PNSQC [11][12][19]. Units: percent of defects present that were found and fixed before release. The four case-study bars come from one documented low-quality case; the inspection and static-analysis values are typical figures from the same analyses; the combined value is the reported best case. The bars are directional, not one controlled experiment.

*Reading Figure 1:* no single test stage removes more than about 40% of defects in the documented case — which is why mature programs layer operations and measure the result end to end as DRE. The published typical DRE near 85% means roughly one defect in seven escapes; that number, tracked over time, is a Level-4 criterion in this model.

**Late discovery is expensive on a measured curve.** Fixing a requirements error cost 1 unit at the requirements phase, 3–8 units at design, and continued escalating through test and operations in a NASA study of three estimation methods [15].

**Factory test governance is now a measurable practice.** A 2026 electronics-production study built test-selection policies with explicit escape-risk bounds over end-of-line datasets — 10,633 training runs, 318 genuine defect events — showing escape-risk quantification is implementable, not aspirational [18].

**Interpretation.** These strands — maturity structures, an outcome case study, DRE economics, cost-escalation curves, and modern factory-test governance — justify both the model existence and its evidence-gating: if maturity did not measurably move defect outcomes, a maturity model would be consulting theater.

## 5.4 What immaturity looks like in practice (composite vignettes, ERC analysis)

Three anonymized composite vignettes — constructed from practitioner experience for illustration, not descriptions of specific companies — make the failure modes concrete. Each maps to specific matrix cells.

**Vignette 1: the bring-up notebook.** A board team brings up a new revision successfully; the engineer who knew the undocumented power-sequencing quirk leaves. The next bring-up stalls for five weeks recreating tribal knowledge. Matrix placement: bring-up L2 (a log exists) failing toward L1 — the artifact is not tied to board revision and no root-cause taxonomy exists.

**Vignette 2: the green suite.** A CI regression suite reports 100% pass on every build. Investigation during diligence shows the suite executes 40 trivial tests, results are HTML screenshots rather than machine-readable records, and no firmware version is bound to any result. Matrix placement: regression automation L2 despite an impressive-looking tool: the results are not reproducible evidence and nothing blocks a merge.

**Vignette 3: the dashboard that never blocked.** A quality dashboard tracks coverage and defect trends. In two years, no release decision ever cites it; releases slip on schedule pressure and the dashboard is presented after the fact. Matrix placement: the measured domains are capped at L2–L3 despite working instrumentation, because gating rule 5 (influence on release decisions) fails — the measurements are decoration.

*Interpretation:* these vignettes illustrate the model central thesis — capability lives in the connection between evidence and decisions, and every gate in Section 6.3 exists to detect its absence.

# 6. The model: levels, domains, and the maturity matrix

## 6.1 The five levels

**Table 6. The five ERC-HVM levels**

| Level | Name | Defining characteristic | Release-decision relationship |
|---|---|---|---|
| L1 | Initial | Ad hoc, manually executed, reactive debugging; success depends on individuals | Release decisions rest on opinion and schedule pressure |
| L2 | Managed | Per-project planning and tracking; basic repeatability at the project level | Release decisions reference a documented, if imperfect, evidence set |
| L3 | Defined | Organization-wide standards, tailored per project; artifacts standardized | Release decisions require defined artifacts; waivers are visible and owned |
| L4 | Quantitatively Managed | Statistical control of selected measures; coverage and DRE tracked against thresholds | Release decisions consume quantitative evidence against thresholds |
| L5 | Optimizing | Continuous improvement from measured data; fault injection, escape analysis, prevention | Release decisions include predicted residual risk and prevention feedback |

(Level logic per CMMI and TMMi [1][3]; the release-decision column is ERC analysis.)

## 6.2 Domain selection and bounding

**(Interpretation)** The 17 domains were selected to cover the full validation lifecycle with minimal overlap, grouped in eight functional bands. Each domain is bounded to avoid double counting: security testing covers firmware-resiliency verification per NIST SP 800-193 but not product certifications; laboratory reproducibility covers measurement validity but not test design.

**Table 7. Domain bounding: what each band includes and excludes**

| Band | Domains | Includes | Excludes (belongs elsewhere) |
|---|---|---|---|
| Inputs | 1 Requirements traceability; 2 Test planning | Requirements-to-test linkage; plan documents and criteria | Requirement authoring quality (product engineering); scheduling |
| Physical execution | 3 Board bring-up; 4 Instrumentation | First power-on through stable testable state; calibrated measurement | Schematic design; fixture mechanical design |
| Firmware-specific | 5 Firmware qualification | Build integrity, signing, manifests, per-module coverage | Firmware feature development; OS bring-up (domain 3) |
| Scale and repeatability | 6 Regression automation; 7 Configuration control | Automated suites, gates, machine-readable results; reproducible configurations | Test-case design technique selection (domain 2); source-code management policy (IT-wide) |
| Defect knowledge | 8 Defect taxonomy and management; 9 Observability | Classification, linkage, detection and diagnostic times | Customer-support ticketing; field service operations |
| Adversarial and stress | 10 Fault injection; 11 Stress testing; 12 Interoperability; 13 Security testing | Deliberate faults, environmental margins, ecosystem compatibility, firmware resiliency | Safety certification campaigns; penetration-test report formats |
| Evidence governance | 14 Test-data governance; 15 Laboratory reproducibility | Schema, retention, integrity, access; run-to-run validity | IT backup policy; general records management |
| Decision closure | 16 Release criteria; 17 Field-feedback integration | Objective release gates and waivers; escape analysis and prevention | Product management go-to-market decisions; warranty policy |

## 6.3 Level rules: the six gates

A domain is scored at level *n* only if **all** of the following hold for level *n* criteria in that domain:

1. The practice is documented and in use (not aspirational).

2. The required artifact exists for the most recent release and is current.

3. The measurement method is stated and reproducible by an assessor from the artifact.

4. A named role owns the practice (individual or team, named in the artifact).

5. The evidence demonstrably influenced at least one release decision in the last cycle (a visible decision record citing the artifact).

6. The evidence carries a confidence label of direct or attested (claimed-only evidence caps the domain at L2 in published profiles).

Tool ownership satisfies none of these alone. A purchased test-management system with empty fields is Level 1 with a license.

## 6.4 The maturity matrix (summary form)

The matrix summarizes, per domain, the distinguishing evidence at each level. Three domains are fully specified per level in Appendix C; Appendix G specifies all 17 domains at Level 3; the assessment workbook carries every domain at full specification for all levels.

**Table 8. Maturity matrix, domains 1–9 (distinguishing evidence per level)**

| # | Domain | L2 | L3 | L4 | L5 |
|---|---|---|---|---|---|
| 1 | Requirements traceability | Requirements list with IDs; one-direction test-to-requirement map | Bidirectional traceability auto-generated from a single source; coverage report per release | Traceability coverage percent tracked against threshold; gaps block release | Traceability data drives test prioritization and prevention feedback |
| 2 | Test planning | Per-project test plan exists and is referenced at review | Organization-standard template with tailoring record; entry and exit criteria defined | Plan adherence measured; exit-criteria pass rates trended | Plans adapt from measured risk and escape data |
| 3 | Board bring-up | Written bring-up log; known-good image per board revision | Checklist tied to board revision and schematic version; blockers logged with root cause | Bring-up duration and blocker taxonomy measured across boards | Bring-up data feeds design-for-testability improvements upstream |
| 4 | Instrumentation | Instrument list; measurements recorded manually | Calibrated instruments with certificates; measurement procedures documented | Measurement uncertainty budgets stated; automated data capture | Uncertainty and coverage jointly optimized; instrument health monitored |
| 5 | Firmware qualification | Build artifact plus smoke test per firmware release | Signed builds verified before execution; version manifest per release | Qualification coverage and defect density per firmware module tracked | Resiliency properties (protect, detect, recover) continuously verified in CI |
| 6 | Regression automation | Some automated tests run on demand | Regression suite runs on every firmware build; failures block merge | Suite runtime, flake rate, and coverage measured with thresholds | Suite optimized by measured defect-detection value |
| 7 | Configuration control | Firmware, driver, and fixture versions recorded per test session | Single source of truth; any historical result reproducible from recorded configuration | Configuration-drift rate measured; non-reproducible results trigger investigation | Configuration data mined to predict flaky or failing configurations |
| 8 | Defect taxonomy and management | Defects logged with severity | Standard taxonomy; every defect linked to failed test and requirement | Detection and containment rates measured by stage | Taxonomy evolved from field data; prevention actions tracked to closure |

| # | Domain | L2 | L3 | L4 | L5 |
|---|---|---|---|---|---|
| 9 | Observability | Serial and console logs saved ad hoc | Standardized telemetry and log schema across benches | Detection time and diagnostic time measured from telemetry | Observability gaps predicted from failure modes before test |

**Table 9. Maturity matrix, domains 10–17 (distinguishing evidence per level)**

| # | Domain | L2 | L3 | L4 | L5 |
|---|---|---|---|---|---|
| 10 | Fault injection | Occasional manual fault tests | Documented injection plan (voltage, clock, bus errors); expected behaviors defined | Injection coverage measured against failure-mode list | Injection automated in CI; results feed FMEA updates |
| 11 | Stress testing | Ad hoc burn-in | Documented stress matrix (temperature, voltage, load) per product | Margin-to-failure measured and trended | Stress results drive derating and design changes quantitatively |
| 12 | Interoperability | Ad hoc device pairing | Documented compatibility matrix; partner-device versions controlled | Interop defect rate per interface tracked | Interop regressions auto-detected from ecosystem telemetry |
| 13 | Security testing | Basic functional security checks | Firmware-update authentication verified (signature checks) per SP 800-193 | Resiliency behaviors (detect, recover) systematically exercised and scored | Continuous adversarial testing; resiliency regressions block release |
| 14 | Test-data governance | Results saved somewhere | Central results store with schema; retention defined | Data quality (completeness, integrity) measured; access controlled | Data reused for prediction; governance audited |
| 15 | Laboratory reproducibility | Results recorded by hand | Environment setups documented; second-run agreement checked | Run-to-run variance quantified against tolerance | Cross-lab correlation program; methods harmonized |
| 16 | Release criteria | Criteria discussed informally | Written release criteria with objective checks; waivers documented and owned | Criteria thresholds statistically justified from measured distributions | Criteria revised from field-return and escape analysis each cycle |
| 17 | Field-feedback integration | Field issues logged | Field defects traced to test gaps; gap-closure actions tracked | Escape root-cause distribution measured; DRE computed | Field data continuously drives prevention and test prioritization |

(Full per-level specifications for every domain are carried in the assessment workbook. All matrix content is ERC analysis anchored to the standards in Table 4.)

## 6.5 Common failure modes at each level (cross-domain, ERC analysis)

**Table 10. Most common failure modes observed at each level**

| Level | Most common failure modes |
|---|---|
| L1 | Heroics; unverifiable "it passed" claims; environment drift between runs; tribal knowledge lost at attrition |

| Level | Most common failure modes |
|---|---|
| L2 | Plans written for audits, not used; traceability maintained by hand and abandoned; bring-up knowledge in personal notebooks |
| L3 | Standards exist but tailoring is fake; artifacts generated after the fact; waivers become routine |
| L4 | Metrics tracked but ignored at release; thresholds set from wishes; DRE computed with short windows that inflate it |
| L5 | Optimization theater: dashboards without decisions; prevention actions logged but never closed |

## 6.6 The five vital-sign domains

**(Interpretation)** A full 17-domain assessment is thorough but expensive. Five domains act as vital signs: if they are weak, the rest of the model is largely decoration, because they gate everything else. The staged assessment (Appendix F) begins with these:

**Table 11. The five vital-sign domains and why they gate the rest**

| Vital-sign domain | Why it gates the others |
|---|---|
| Requirements traceability (1) | Without it, regression results do not connect to obligations; every coverage claim is unanchored |
| Regression automation (6) | Without machine-readable, version-bound results, no other domain can be measured over time |
| Configuration control (7) | Without recorded configurations, failures cannot be reproduced — the core capability the model exists to assess |
| Release criteria (16) | Without objective gates, evidence produced by every other domain has no decision to influence |
| Defect taxonomy and management (8) | Without linked, classified defects, prevention and escape analysis (Levels 4–5) are impossible |

An organization at L2–L3 across the five vital signs is a functioning, improvable organization; one at L1 on all five is carrying unquantifiable release risk regardless of how strong its engineers are individually.

# 7. Scoring rules and anti-precision constraints

## 7.1 Scoring rules

**(Calculated result — rules stated; the arithmetic is inspectable in the workbook.)**

**1. Domain score.** The domain score is the highest level whose criteria are all evidenced per the gating rules. Partial credit is not awarded; the workbook records evidenced or not-evidenced per criterion, and the level follows mechanically.

**2. Profile over number.** The assessment output is a 17-domain vector displayed as a heat map (Appendix D). A single overall level may be stated only as the median domain level with its distribution shown — for example, "median L2; range L1–L4; see heat map" — never as an average decimal.

**3. No unsupported precision.** Levels are ordinals. Arithmetic implying "2.7" or "L2.5" is prohibited in deliverables derived from this model. Trends are reported as counts of domains at each level over time.

**4. Confidence labels.** Every domain score carries an evidence-confidence label: direct (assessor inspected artifacts), attested (owner demonstration), or claimed (asserted, artifact not shown). Claimed evidence caps the domain at L2 in published profiles.

**5. Freshness.** Evidence older than one release cycle, or twelve months, expires for scoring purposes.

**6. Conflict rule.** If an artifact and an interview contradict each other, the artifact governs and the discrepancy is recorded.

## 7.2 Why tool ownership earns nothing

Assessors commonly inflate scores by counting licenses: a test-management suite, a CI system, a boundary-scan tool. This model gates on whether processes produce repeatable evidence and influence release decisions. The operative questions are: "Show me the last release regression results with their recorded firmware versions and thresholds," and "Show me the release decision that cited them." A modest open-source setup with disciplined evidence outscores a large toolchain without it. This gating is consistent with the evidence-based logic of the CMMI and TMMi frameworks [3][4]. (ERC analysis.)

## 7.3 Worked scoring example (illustrative, ERC-constructed)

**(Illustrative example constructed by ERC for exposition; not a description of any real organization.)** Consider a mid-size controls manufacturer assessed on regression automation. The workbook records, per criterion:

**Table 12. Worked example: regression automation domain scoring**

| Criterion set | Evidenced? | Evidence examined | Confidence |
|---|---|---|---|
| L2: automated tests exist and run on demand | Yes | Scripts plus a run log for the last three releases | Direct |
| L3: suite runs on every firmware build; failures block merge; results machine-readable with recorded versions | Yes | CI configuration; JSON results binding firmware build IDs; one merge blocked on a failing run, per the gate log | Direct |
| L3: measurement method stated | Partial | No documented method for extracting metrics; assessor could not reproduce counts from the artifact | — |
| L4: runtime, flake rate, and coverage measured against thresholds | No | Flake rate discussed in meetings; no threshold defined; no report artifact | — |
| Gate 5: evidence influenced a release decision | Yes | Release-decision record cites the CI gate report for the last two releases | Direct |

**Assignment.** L3 criteria are met except the measurement-method criterion (gate 3 fails at L3); L4 is unevidenced. The domain therefore scores **L2 with direct confidence**, with the remediation note: "document the metric-extraction method to secure L3; define flake-rate

and coverage thresholds to reach L4." The workbook performs exactly this arithmetic; nothing else is computed.

### 7.4 Executive read-out format

Executive deliverables state: the median level with its distribution; the three lowest-scoring domains with schedule-risk and escape-risk consequences in plain language; the estimated remediation sequence; and the evidence that would change the assessment. No decimal scores and no letter grades appear in executive or any other deliverables.

## 8. Process maturity versus product quality

**Governing point (Interpretation).** A mature process is not a defect-free product, and a defect-free quarter is not a mature process. Conflating them produces both false confidence and false blame.

**Process maturity** measures whether the organization can reliably find, reproduce, prioritize, and prevent defects: traceability, controlled configurations, regression evidence, measured thresholds. **Product quality** measures what actually shipped: escape counts, field returns, DRE.

The model Level-4 and Level-5 criteria consume product measures as inputs (DRE, escape root-cause distribution) while the score itself still rests on process evidence. An L4 organization that discovers a serious defect is functioning correctly: its evidence shows when the defect was introduced, why coverage missed it, and what changed. An L1 organization discovering the same defect can answer none of those questions.

**Worked DRE example (Calculated result — illustrative numbers constructed by ERC).** Suppose a program found and fixed 1,200 defects before release, and field testing over twelve months confirmed 180 additional defects attributable to the shipped firmware. DRE = 1,200 / (1,200 + 180) = 87.0%. Two cautions from the literature apply: a 30-day observation window would yield an artificially high DRE because early field exposure is thin [12][13]; and DRE must count only confirmed field defects, not support tickets. Tracked against a threshold over successive releases, this single number summarizes the combined effect of every domain in the model.

This separation also protects the model from the two classic abuses: claiming maturity because a quarter had no escapes, and condemning a process because a hard defect appeared despite mature evidence showing exactly where the process caught it late.

## 9. Risks, limitations, uncertainty, and data gaps

**Lead point.** The most consequential limitation is that the model has not yet been interview-validated; that validation is scheduled and its results will be published with any revisions.

**Table 13. Limitations, consequences, and mitigations**

| Limitation | Consequence | Mitigation |
|---|---|---|
| Model validity not yet interview-validated | Criteria may misfit some industries | Structured practitioner interviews scheduled before final release; revisions published |
| Domain scores can be gamed (artifact theater) | Inflated levels | Gating rule 5 (evidence must have influenced a release decision); assessor reproduces the measurement from the artifact |
| Ordinal levels invite false precision | Overconfident decisions | Anti-precision rules (Section 7.1); heat-map profile required in all deliverables |
| Seventeen domains by five levels is a large assessment | Cost and time | Workbook supports staged assessment: five vital-sign domains first (Section 6.6) |
| Industry variation (semiconductor versus small OEM) | The same evidence means different effort | Tailoring guidance by organization type and size (Appendix H); scale noted in every published profile |
| DRE and escape data depend on field visibility | Level 4/5 evidence is weak where telemetry is absent | Alternative criteria paths defined in the workbook for constrained products |
| The model is ERC-developed | Potential self-interest bias | Open publication; instruments usable without ERC; commercial interest disclosed |
| Standards cited are paywalled (ISO) | Readers cannot verify clause text | Citations identify standards and clause areas; no quotation of paywalled text |
| Anchor frameworks evolve (CMMI, TMMi versions) | Criterion wording may lag framework revisions | Model versioning tracks referenced framework editions; change log updated on revision |

## 9.1 Quality assurance gates applied to this report

**Editorial review.** Title matches scope; the executive summary agrees with the body; every research question is answered or explicitly unresolved; conclusions stay within the evidence; the model-derivation versus external-evidence distinction is labeled throughout; neutral naming throughout; acronyms defined at first use.

**Source audit.** Every link cited was opened and checked against the exact claim during preparation (access dates October 27, 2026). Framework and government sources are primary; DRE analyses are practitioner benchmarks with stated methods, used as the best available data and never as controlled experiments; the single case study is labeled as such everywhere it appears.

# 10. Adoption roadmap and remediation template

## 10.1 Sequenced adoption

**(Recommendation)** Improvement follows the documented evolutionary logic of maturity frameworks — each level builds on the prior plateau [4] — applied to the vital-sign domains first.

**Table 14. Sequenced adoption roadmap**

| Step | Target | Typical actions | Evidence of completion |
|---|---|---|---|
| 1 (weeks 1–4) | Baseline | Self-assess the five vital-sign domains with the workbook; assemble the evidence pack | Completed workbook rows with confidence labels |

| Step | Target | Typical actions | Evidence of completion |
|---|---|---|---|
| 2 (months 1–3) | L2 on vital signs | Written release criteria; defect log with severity; configuration capture per test session | Artifacts in use at the last release |
| 3 (months 3–9) | L3 on vital signs | Bidirectional traceability auto-generated; regression suite on every build; standardized results schema | Traceability report per release; CI gate evidence |
| 4 (months 9–18) | L4 on vital signs | Coverage and DRE thresholds; configuration-drift and flake-rate measurement; statistical exit criteria | Thresholds with distributional justification |
| 5 (continuous) | L5 selectively | Fault injection in CI; escape root-cause feedback; prevention tracking to closure | Prevention actions closed with measured effect |

(Timing bands are ERC estimates; the ordering follows the TMMi documented evolutionary path [4].)

### 10.2 Remediation roadmap template (per domain)

For each below-target domain, the roadmap records: the current level and confidence label; the specific unmet criteria; the artifact to be created or fixed; the owning role; the due release; the verification method (how the assessor will re-check); and the dependency chain on other domains — for example, regression automation at L3 depends on configuration control at L3, because failures cannot be reproduced without recorded configurations.

### 10.3 Due-diligence read-out template

For investors and acquirers: the domain heat map; the five vital-sign levels; the top three risks translated into schedule, escape, and rework language; the questions to ask management (drawn from the interview guide in Appendix B); and the evidence requests whose refusal is itself a finding. ERC will not publish ratings of named organizations; diligence reports are client-confidential.

## 11. Conclusion

**Conclusion.** Capability in hardware and firmware validation is describable, assessable, and improvable — provided the model gates on evidence and release influence rather than tool ownership, and reports profiles instead of false precision. ERC-HVM 1.0 provides that structure: five levels, 17 domains, every criterion specified with expected practice, required artifact, measurement method, responsible role, and common failure modes; scoring rules that resist inflation; and instruments — evidence-request list, interview guide, workbook, heat map — published for anyone to use. Edition 2 adds interview-validated criteria and the first worked assessments. For ERC, this is the reusable methodology behind future paid assessments — published openly, on purpose, so the methodology quality is always inspectable.

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Access dates: October 27, 2026. The final release will include archived snapshots where permitted.


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*Note on source quality:* references 1–4 and 20 are authoritative framework publications; references 5–10 are primary government and international standards; references 11–14, 17, and 19 are widely cited practitioner analyses with stated methods — the best available benchmark data, not controlled experiments; reference 16 is a peer-reviewed single case study, labeled as such wherever cited; reference 18 is a peer-reviewed 2026 study; references 21–22 are vendor or secondary materials used only to characterize practice, never as the sole basis for a material claim.

# Appendix A: Evidence-request list

## A.1 Universal requests (all domains)

The last release decision record; the artifacts cited in it; the configuration record for the most recent regression run; the defect-database export with taxonomy fields; and the calibration certificates for instruments used in the last release cycle.

## A.2 Domain-specific requests (abridged; the workbook carries the full list)

• **Requirements traceability:** the auto-generated bidirectional traceability report for the last release; the coverage-gap list; the release decision citing it.

• **Regression automation:** CI run history for the last 30 builds; one failing run recorded preconditions, firmware and driver versions, and environment; the flake-rate report.

• **Firmware qualification:** the signed build manifest; verification logs showing signature or hash checks before execution; the version-to-board-revision compatibility matrix.

• **Fault injection:** the injection plan; the expected-behavior table; the last executed injection results.

• **Security testing:** SP 800-193-style protect, detect, and recover verification evidence; firmware-update authentication test results.

• **Laboratory reproducibility:** documented environment setup; run-to-run agreement data; the calibration traceability chain.

• **Field-feedback integration:** the escape root-cause distribution for the last two releases; gap-closure actions and their status.

Refusing, or being unable, to produce these items is itself scored evidence.

# Appendix B: Interview guide (structured, 60–90 minutes)

**Opening (5 minutes):** scope, confidentiality, the evidence-over-opinion framing, and recording consent.

**Per-domain prompts (six to eight domains selected per session):**

• "Walk me through what happens when a board fails bring-up. Where does the log go? Who sees it?"

• "Show me how you would reproduce last Tuesday regression failure on a different bench."

• "Which firmware version, driver version, and fixture were in play for the last release regression run? Where is that recorded?"

• "When release criteria were last waived, who decided, and where is that written?"

• "How did the last field return change your test suite?"

• "What threshold would block a release today, and where did that number come from?"

• "Which of your tests could a new engineer run unaided? Which require one specific person?"

**Cross-checks:** answers are cross-checked against the artifacts in Appendix A; contradictions are recorded under the conflict rule (Section 7.1).

**Closing:** "If you could fix one thing in validation this year, what and why?" — feeds the remediation-roadmap plausibility check.

# Appendix C: Three fully specified exemplar domains

## C.1 Requirements traceability

**Table 15. Requirements traceability — full specification per level**

| Field | L2 | L3 | L4 | L5 |
|---|---|---|---|---|
| Expected practice | Test cases map to requirement IDs, maintained manually | Bidirectional traceability auto-generated from a single source; gaps reported per release (NPR 7150.2D SWE-052 alignment) | Coverage percent against threshold; unresolved gaps block release | Traceability data prioritizes testing by risk; gaps feed prevention |
| Required artifact | Requirements list plus test-to-requirement map | Generated bidirectional report per release; gap list | Coverage trend report with threshold | Prioritization report plus prevention log |
| Measurement method | Manual count | Script-generated counts from the source system | Scripted count versus threshold; trend over at least two releases | Same, plus measured effect of prioritization on the escape distribution |
| Responsible role | Test lead | Requirements engineer and test lead | Quality manager | Quality manager and chief engineer |
| Common failure modes | The map rots after the first update; IDs renamed without propagation | Generation script undocumented; the "single source" is three files | Threshold set arbitrarily; gaps waived routinely | Prioritization exists, but escape data never closes the loop |

## C.2 Regression automation

**Table 16. Regression automation — full specification per level**

| Field | L2 | L3 | L4 | L5 |
|---|---|---|---|---|
| Expected practice | Automated tests exist and run on demand | Suite runs on every firmware build; failure blocks merge; results machine-readable with recorded versions | Runtime, flake rate, and coverage measured against thresholds | Suite composition optimized by measured defect-detection value; redundant tests retired |

| Field | L2 | L3 | L4 | L5 |
|---|---|---|---|---|
| Required artifact | Test scripts and a run log | CI configuration; machine-readable results with metadata | Metrics report per release; threshold definitions | Test-value analysis; retirement log |
| Measurement method | Manual run, manual read | Automatic execution and result capture | Scripted metric extraction from the results store | Detection-value model with stated assumptions |
| Responsible role | Test engineer | Build/CI owner and test lead | Quality manager | Quality manager and architects |
| Common failure modes | The suite passes because it tests little; results unreadable by machines; flaky tests disabled silently | CI green despite unrecorded environment; the merge block is bypassed by override | Metrics collected but not consulted at release | Optimization removes tests that were the only coverage of a failure mode |

## C.3 Firmware qualification

**Table 17. Firmware qualification — full specification per level**

| Field | L2 | L3 | L4 | L5 |
|---|---|---|---|---|
| Expected practice | Build artifact identified; smoke test run | Builds signed and verified before execution; release manifest lists firmware versions per board revision (NIST SP 800-193 alignment) | Per-module qualification coverage and defect density tracked with thresholds | Protect, detect, and recover resiliency behaviors verified continuously in CI |
| Required artifact | Build log plus smoke result | Signed image; verification log; manifest | Coverage and defect-density report by module | Continuous resiliency-verification results |
| Measurement method | Manual inspection | Automated signature verification; manifest check | Scripted aggregation from the results store | CI-integrated injection and verification |
| Responsible role | Firmware engineer | Release engineer and security engineer | Quality manager | Security architect |
| Common failure modes | "It boots" equals qualified; unsigned test images ship; manifest stale | Signature checks skipped in debug builds that reach the field | Module coverage gamed by trivial tests | Resiliency tests exist, but findings do not block release |

The remaining 14 domains receive the same five-field, per-level specification in the assessment workbook; publishing them in the workbook keeps this report readable while keeping every criterion inspectable.

# Appendix D: Heat-map template

The heat map is a table with one row per domain. Structure per row: domain name; five level cells (L1–L5); an evidence-confidence column (direct / attested / claimed); a decisive-artifact reference column; and a note column. Rules: the attained level is marked with a filled cell carrying a distinct texture or symbol in addition to shading — never color alone; every row

cites the artifact reference that justified its level; rows whose evidence is claimed-only are visually distinguished and capped at L2 in published profiles. The heat map is reproduced as a plain table in every deliverable so the profile is readable without any graphic.

# Appendix E: Accessibility conformance statement

**Target.** WCAG 2.2 Level AA plus PDF/UA as the internal ERC standard for public reports.

**Implemented in this edition.** Letter page size; 0.75-inch margins; 10.5 pt body text left-aligned (not justified); headings implemented as distinct paragraph styles in a two-to-three-level hierarchy (a fourth level is not used); consistent headers and footers carrying the report identifier, version, and page numbers; Unicode-mapped searchable text; live hyperlinks with descriptive text and PDF bookmarks matching the table of contents; document title, author, subject, and keyword metadata set; tables with header rows and grid borders (no layout-only tables), headers repeated across pages; heat-map marks use texture plus shading rather than color alone; minimum 4.5:1 text contrast. One typographic deviation is recorded: third-level headings are set at 11.5 pt bold oblique to preserve visual hierarchy below the 14 pt second-level headings.

**Deviations and planned remediation.** This edition is generated from a programmatic layout engine that does not yet emit a complete PDF/UA tag tree. Before final release, the PDF will be remediated and validated per the ERC accessibility acceptance process: automated checker, PDF/UA validation (PAC or equivalent), manual tag-tree and reading-order inspection, keyboard and screen-reader sample testing, plain-text extraction review, and a recorded test log. The accessible HTML edition is the canonical reading experience; this PDF serves offline, print, and archival use. An alternate-format contact appears in the document-control section.

**Figures and tables.** Figure 1 presents every plotted value as text adjacent to its bar; its complete underlying data appear in the source note and Section 5.3. All tables use real header rows, repeated across pages.

# Appendix F: Assessment operations and workbook

## F.1 Workbook contents (machine-readable)

**Table 18. Assessment workbook structure**

| Sheet | Contents |
|---|---|
| 1 — Criteria | 17 domains by 5 levels; every criterion with its five fields: expected practice, required artifact, measurement method, responsible role, common failure modes |
| 2 — Assessment record | Evidenced / not-evidenced per criterion; confidence labels; artifact references; dates |
| 3 — Profile | Computed domain levels; heat-map data; median and distribution (counts only) |
| 4 — Remediation roadmap | Per-domain template with owners, due releases, verification methods, dependencies |

| Sheet | Contents |
|---|---|
| 5 — Decision citation log | Release decisions that cited each evidence item, per the gating rule |

All formulas are visible; nothing is hidden; the workbook computes nothing beyond level assignment and counts.

### F.2 Assessment flow and effort

**(Recommendation)** Evidence pack requested (Appendix A) → artifact review → structured interview (Appendix B) → scoring under the conflict rule → heat map and executive read-out (Section 7.4) → remediation roadmap (Section 10.2). Typical effort: 3–5 assessor-days for the vital-sign scope; 8–12 assessor-days for the full 17 domains, depending on organization size. Every published engagement discloses whether equipment, data access, or compensation shaped the assessment conditions.

## Appendix G: Level-3 (Defined) criteria for all 17 domains

Level 3 is the pivotal rung: it converts project-level habits into organization-wide, evidence-producing standards. The table states, per domain, the expected practice, required artifact, measurement method, responsible role, and the characteristic failure mode at L3. Full per-level specifications for every level are in the workbook.

**Table 19. All domains at Level 3 (Defined)**

| Domain | Expected practice at L3 | Required artifact | Measurement method | Responsible role | Characteristic L3 failure mode |
|---|---|---|---|---|---|
| 1 Requirements traceability | Bidirectional traceability auto-generated from a single source; gaps reported per release | Generated bidirectional report; gap list | Script-generated counts from the source system | Requirements engineer + test lead | Generation script undocumented; "single source" is three files |
| 2 Test planning | Organization-standard plan template with tailoring record; entry and exit criteria defined | Tailored plan per project | Template-compliance review | Test lead | Tailoring record written after the fact |
| 3 Board bring-up | Bring-up checklist tied to board revision and schematic version; blockers logged with root cause | Revision-bound bring-up checklist and log | Checklist-completion audit | Hardware lead | Checklist exists but is bypassed under schedule pressure |
| 4 Instrumentation | Calibrated instruments with certificates; measurement procedures documented | Calibration certificates; procedure documents | Certificate currency check | Lab manager | Certificates lapsed; procedures unread |
| 5 Firmware qualification | Signed builds verified before execution; version manifest per release | Signed image; verification log; manifest | Automated signature verification; manifest check | Release engineer | Debug builds without signature checks reach the field |

| Domain | Expected practice at L3 | Required artifact | Measurement method | Responsible role | Characteristic L3 failure mode |
|---|---|---|---|---|---|
| 6 Regression automation | Regression suite runs on every firmware build; failures block merge; results machine-readable | CI configuration; machine-readable results | Automatic execution and capture | Build/CI owner | Merge block bypassed by override |
| 7 Configuration control | Single source of truth; any historical result reproducible from recorded configuration | Configuration record per session | Reproduction attempt by assessor | Test lead | Record exists but omits fixture and instrument versions |
| 8 Defect taxonomy and management | Standard taxonomy; every defect linked to failed test and requirement | Defect export with taxonomy and linkage fields | Linkage-completeness query | Defect manager | Links added retrospectively before audits |
| 9 Observability | Standardized telemetry and log schema across benches | Telemetry schema document; sample captures | Schema-conformance check | Firmware architect | Schema defined but benches emit legacy formats |
| 10 Fault injection | Documented injection plan (voltage, clock, bus errors); expected behaviors defined | Injection plan; expected-behavior table | Plan-completeness review | Validation engineer | Expected behaviors undefined, so any result passes |
| 11 Stress testing | Documented stress matrix (temperature, voltage, load) per product | Stress matrix document | Matrix-completeness review | Validation engineer | Matrix never updated for new hardware revisions |
| 12 Interoperability | Documented compatibility matrix; partner-device versions controlled | Compatibility matrix | Matrix-currency audit | Integration lead | Partner versions uncontrolled, making failures irreproducible |
| 13 Security testing | Firmware-update authentication verified (signature checks) per SP 800-193 | Authentication test results | Automated verification | Security engineer | Tests exist only for the happy path |
| 14 Test-data governance | Central results store with schema; retention defined | Schema document; retention policy | Schema-conformance and retention checks | Data owner | Store exists; many results arrive out of schema |
| 15 Laboratory reproducibility | Environment setups documented; second-run agreement checked | Environment document; agreement data | Second-run comparison | Lab manager | Agreement checked only when results look suspicious |
| 16 Release criteria | Written release criteria with objective checks; waivers documented and owned | Release-criteria document; waiver log | Criteria-execution audit | Release board | Waivers become the normal path |
| 17 Field-feedback integration | Field defects traced to test gaps; gap-closure actions tracked | Escape analysis report; action log | Action-closure audit | Quality manager | Analysis written; actions never assigned |

# Appendix H: Tailoring by organization type (ERC analysis)

The same evidence means different effort and takes different shapes across organization types. The model does not relax criteria by size — a smaller organization can absolutely reach L3 — but assessors should expect the artifacts to look different, and should score what exists, not what a large company would produce.

**Table 20. Expected artifact shapes and typical first gaps by organization type**

| Organization type | What L3 evidence typically looks like | Typical first gaps |
|---|---|---|
| Semiconductor company (silicon or SiP) | Formal per-silicon qualification plans; ATE regression suites; versioned test programs tied to design revisions; resiliency verification for boot firmware | Fault-injection coverage against the failure-mode list; field-escape feedback into ATE prioritization |
| Board and system manufacturer | Eco- and revision-bound bring-up checklists; boundary-scan and functional test programs under configuration control; signed firmware manifests | Instrument-calibration discipline; bidirectional traceability from requirements to functional test |
| Embedded-device company / OEM | Per-product stress and interop matrices; firmware-update authentication tests; standardized log schemas across products | Test-data governance (schema discipline); release criteria with statistical justification |
| Small OEM / startup | Lean but real: single-source configuration file, scripted traceability, CI gate on builds — achievable without headcount | Everything is in one engineer head; regression automation absent; release criteria informal |
| Test laboratory | ISO/IEC 17025-aligned procedures; calibration traceability; documented uncertainty budgets | Reproducibility as a measured metric (run-to-run variance vs tolerance); automated data capture |
| Workforce-development program | Curriculum artifacts mapped to model criteria; student-run benches with configuration and evidence discipline | Sustained practice beyond course scopes; defect-linkage habits |

# Appendix I: Scoring quick-reference card

A one-page summary of the rules an assessor must apply, suitable for printing and for inclusion in engagement kick-off materials.

**Table 21. Scoring quick reference**

| Rule | Statement |
|---|---|
| Six gates (per level, per domain) | 1 practice documented and in use; 2 artifact current; 3 measurement reproducible; 4 named role owns it; 5 evidence influenced a release decision; 6 confidence label direct or attested |
| Domain score | Highest level whose criteria are all evidenced; no partial credit |
| Profile | 17-domain heat map is the deliverable; single overall level only as median with distribution shown |
| No decimals | Levels are ordinals; "2.7" and "L2.5" are prohibited everywhere |
| Confidence caps | Claimed-only evidence caps the domain at L2 in published profiles |
| Freshness | Evidence older than one release cycle or twelve months expires |
| Conflict rule | Artifact governs over interview; discrepancies recorded |
| Tool rule | Tool ownership satisfies no gate by itself |

| Rule | Statement |
|---|---|
| Executive format | Median level, three weakest domains in plain risk language, remediation sequence, and what evidence would change the result |

# Appendix J: Level-2 floor criteria for all 17 domains

Level 2 is the floor of managed practice: if an organization cannot evidence these items, it is Level 1 in that domain regardless of sophistication elsewhere. The L2 floor doubles as a minimum onboarding checklist for new validation teams and workforce programs.

**Table 22. Level-2 (Managed) floor criteria for all 17 domains**

| Domain | L2 expected practice | L2 required artifact |
|---|---|---|
| 1 Requirements traceability | Test cases map to requirement IDs, maintained manually | Requirements list; test-to-requirement map |
| 2 Test planning | Per-project test plan exists and is referenced at review | Test plan document |
| 3 Board bring-up | Written bring-up log; known-good image per board revision | Bring-up log; known-good image reference |
| 4 Instrumentation | Instrument list; measurements recorded manually | Instrument list; manual measurement records |
| 5 Firmware qualification | Build artifact plus smoke test per firmware release | Build log; smoke-test result |
| 6 Regression automation | Some automated tests run on demand | Test scripts; run log |
| 7 Configuration control | Firmware, driver, and fixture versions recorded per test session | Per-session version record |
| 8 Defect taxonomy and management | Defects logged with severity | Defect log with severity field |
| 9 Observability | Serial and console logs saved ad hoc | Saved log files |
| 10 Fault injection | Occasional manual fault tests | Ad hoc test notes |
| 11 Stress testing | Ad hoc burn-in | Burn-in notes |
| 12 Interoperability | Ad hoc device pairing | Pairing notes |
| 13 Security testing | Basic functional security checks | Security check results |
| 14 Test-data governance | Results saved somewhere | Stored result files |
| 15 Laboratory reproducibility | Results recorded by hand | Hand-recorded results |
| 16 Release criteria | Criteria discussed informally | Meeting notes or email threads |
| 17 Field-feedback integration | Field issues logged | Field-issue log |

Every L2 row in this table is the assessment workbook starting point: an organization that cannot check these 17 rows has its roadmap already — build the floor before anything else.

# Glossary and acronyms

**Bidirectional traceability** — associations discernible in either direction between logical entities (NASA and ISO vocabulary). **Bring-up** — first power-on and incremental enabling of a board or system. **CI** — continuous integration. **CMMI** — Capability Maturity Model Integration (ISACA). **DRE** — defect removal efficiency: defects found and fixed before release divided by defects present. **Escaped defect** — a defect discovered after handoff by a party other than the developing organization. **FMEA** — failure mode and effects analysis. **HVM** — the ERC Hardware and Firmware Validation Maturity Model. **ISO/IEC 17025** — the laboratory-competence standard. **ML / CL** — maturity level / capability level. **NPR** — NASA Procedural Requirement. **RTU / RTD / CTD** — roots of trust for update, detection, and recovery (NIST SP 800-193 terms). **SP** — NIST Special Publication. **SSDF** — Secure Software Development Framework. **TEVV** — test, evaluation, verification, and validation. **TMMi** — Test Maturity Model integration. **Validation** — confirmation through objective evidence that requirements for a specific intended use are fulfilled. **Wilson interval** — a confidence interval for proportions with better small-sample behavior than the normal approximation.

# Change log

**Table 23. Version history**

| Version | Date | Change |
|---|---|---|
| 1.0 | 2026-10-27 | Initial public draft: model definition; 17-domain maturity matrix; scoring and anti-precision rules; evidence-request list; interview guide; exemplar domains; Level-3 criteria for all domains; tailoring guidance; scoring quick-reference; L2 floor criteria. |
| 1.0a | 2026-10-28 | Production remediation: embedded TrueType typefaces; document language (en-US) declared in the PDF catalog; accessibility pre-test record published; companion accessible HTML edition generated. |
| 1.1 (planned) | 2027 Q1 | Practitioner-interview validation results; criteria revisions; industry tailoring notes. |
| 2.0 (planned) | 2027 | First worked assessments published as anonymized case studies; workbook v2. |

# Data and code statement

The assessment workbook (XLSX/CSV), heat-map template, interview guide, and evidence-request list will be published on the ERC canonical report page for ERC-RPT-2026-003 under CC BY 4.0, with any scoring scripts under an OSI-approved license. This edition contains no organization-specific data. Interview recordings and assessment records from future engagements remain confidential to clients; only anonymized, aggregated statistics will be published, and only with client consent.

**Corrections.** Corrections, replacement versions, the workbook, and supporting files will be posted on the canonical ERC report page identified in the document-control section. Readers who identify an error may contact ERC through that page.